25901122 **ANSWER KEY**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEBOTA01 - AGRIFOOD SCIENCE AND NANOTECHNOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short answer - Answer any 10 questions
Each question carries 2 marks

[2x10 = 20]

1. Name two nanomaterials used in agriculture. **[K] / [CO1]**
Nano-ZnO, Nano-Fe_2O_3 (Nano – Zinc Oxide, Nano Ferric Oxide/ Iron Oxide) (any two).
These improve enzyme activity, chlorophyll content, and plant metabolism.

2. List two activities involved in agrifood systems. **[K] / [CO1]**
Food production, Food processing
Also includes storage, distribution, and marketing (any two).

3. Name two examples of physical synthesis techniques. **[K] / [CO1]**
Laser ablation, Arc discharge method - These produce high-purity nanoparticles without chemical contamination (any two).

4. List two bio-nanomaterials. **[K] / [CO1]**
Chitosan nanoparticles, protein nanoparticles
These are biodegradable and eco-friendly

5. What are food additives? Give two examples. **[U] / [CO2]**
Preservatives, Flavor enhancers - also include colorants and antioxidants (any two).

6. What is the role of gold nanoparticles in nanosensor kits? **[U] / [CO3]**
Used in nanosensors, provide colorimetric detection, enhance sensitivity
Color change indicates pathogen presence.

7. What are nanosensors? **[U] / [CO2]**
Devices using nanomaterials to detect: chemical changes, biological contaminants. physical parameters - Eg: Detect *E. coli* in food.

8. What are RFID-based nanotags used in smart packaging? **[U] / [CO3]**
RFID-based nanotags: Store product data, enable tracking, ensure traceability
Used in: Supply chain monitoring.

9. How do nano-fertilizers impact a crop's photosynthetic efficiency? [U] / [CO3]
Improve nutrient absorption, enhance chlorophyll synthesis, increase photosynthetic rate.

10. How does nanotechnology help reduce food wastage? [U] / [CO4]
Smart packaging, spoilage detection, extended shelf life

11.What is the importance of integrating nanotechnology with digital tools in modern agrifood systems? [U] / [CO4]
Real-time monitoring, AI-based decisions, precision agriculture

12. How does nanotechnology improve crop productivity? [U] / [CO3]
Controlled nutrient release, reduced losses, better plant growth

Part B
Short essays - Answer any 8 questions
Each question carries 5 marks

[5x8 = 40]

13. Describe food products obtained from aquaculture. [K] / [CO1]
Fish (salmon, tilapia), Shellfish (shrimp, crab), Seaweed, Fish oil - rich in protein, omega-3 fatty acids, and minerals, edible aquatic plants – Lotus – rich in carbohydrates and fibres
(any 5 with a brief description)

14. Describe the method of encapsulating vitamins or antioxidants using nanoliposomes.
[U] / [CO2]
Phospholipid bilayer vesicles - Hydrophilic core + hydrophobic shell
Explain the process: process flowchart in a nutshell:

Bioactive compound
↓
Mix with phospholipids
↓
Formation of bilayer vesicle
↓
Encapsulation inside liposome
↓
Stable delivery system

15. Describe the different types of nanobarcodes used in food packaging systems.
[U] / [CO3]
Types: Optical nanobarcodes, Magnetic nanobarcodes (brief explanation needed)

Food product
↓
Nanobarcode tagging
↓
Data storage (origin, expiry)
↓
Scanning & tracking

Importance: Prevents fraud and ensures traceability

16. Explain the benefits of nanofiltration compared to traditional filtration and thermal processing methods in fruit juice industries. **[U] / [CO3]**

Advantages: Removes microbes & toxins, retains nutrients, no heat damage

In a nutshell:

Raw juice
↓
Nanofiltration membrane
↓
Removal of contaminants
↓
Pure, nutrient-rich juice

17. Explain the procedure for preparing plant extract used in green synthesis of silver nanoparticles.

[U] / [CO4]

Steps: Wash leaves- Cut- Boil- Filter *(process in brief)*

Plant leaves
↓
Washing
↓
Boiling (10–15 min)/ extraction with solvents or water
↓
Filtration
↓
Plant extract ready

18. How protein nanoparticles maintain the quality of packaged food? **[U] / [CO2]**

Role: Encapsulation of nutrients, controlled release, protection from oxidation

19. Describe how clear regulations are very important for the development of nanobased agricultural products?

[U] / [CO3]

Safety assurance
Toxicity control
Standardization
Consumer trust

Flow: Research → Testing → Regulation → Approval → Commercial use

20.Write a short note on the role of IITs in nanoscience and nanotechnology research.

[K] / [CO1]

Development of nanosensors, smart agriculture tools, research in nanomaterials
Eg: Soil nutrient sensors.

21.What are three environmental issues from conventional fertilizers that traditional fertilizers are proven to lessen? **[K] / [CO1]**

Nutrient leaching, Soil pollution, Water contamination
Additional: Reduced greenhouse gas emissions.

(LIBERAL VALUATION) – If attempted can be given full marks

22. Explain the role of nano-enabled "Smart Hydroponics" in urban settings. [K] / [CO1]

Features: Nano-nutrients, Sensors, Controlled environment

In brief

Water + Nutrients

↓

Nano-form nutrients

↓

Plant root absorption

↓

Sensor monitoring

↓

Optimized growth

Part C

Essays

Answer any 1 question. Each question carries 10 marks

[10x1 = 10]

23. Outline the major applications of nanotechnology in different fields. [K] / [CO1]

Nanotechnology applications: Agriculture, Food , Medicine, Environment, Electronics

1. Agriculture - Nano-fertilizers, nano-pesticides, sensors
2. Food Industry - Nanoencapsulation, flavor control
3. Packaging- Smart packaging, antimicrobial films
4. Medicine - Drug delivery, Imaging
5. Environment -Water purification
6. Electronics -Quantum dots, Nanochips

24. Discuss the potential toxic effects of nanomaterials on soil health, plant systems, and human health within agricultural ecosystems. [U] / [CO3]

1. Soil Health - Microbial disruption, Reduced fertility
2. Plant Systems - Oxidative stress, DNA damage
3. Human Health – Bioaccumulation, Toxicity risks
4. Environmental Impact

(In brief)

Nanoparticles

↓

Soil/Water contamination

↓

Plant uptake

↓

Food chain transfer

↓

Human exposure
